

3(a)	$(m \times 3.0) \text{ or } (2.5 \times 9.6 \times \cos 60^\circ)$	C1
	$(m \times 3.0) - (2.5 \times 9.6 \times \cos 60^\circ) = 0 \text{ so } m = 4.0 \text{ (kg)}$	A1

Question	Answer	Marks
3(b)	$2.5 \times 9.6 \times \sin 60^\circ = (4.0 + 2.5) \times V$	C1
	$V = 3.2 \text{ m s}^{-1}$	A1
	or use of momentum vector triangle: $(4.0 \times 3.0)^2 + [(4.0 + 2.5) \times V]^2 = (2.5 \times 9.6)^2$	(C1)
	$V = 3.2 \text{ m s}^{-1}$	(A1)
3(c)	$E = \frac{1}{2}mv^2$	C1
	difference in $E_K = \frac{1}{2} \times 2.5 \times (9.6)^2 - \frac{1}{2} \times 4.0 \times (3.0)^2$	
	$= 97 \text{ J}$	A1

Question	Answer	Marks
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